Alex Zongo

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RESEARCH FOCUS

Safe decision-making for multi-agent autonomous systems: robust multi-agent reinforcement learning under degraded or adversarial sensing, runtime safety filtering in learned control policies, and foundation models in the decision loop, with air traffic control for both manned and unmanned aircraft as the proving ground.

EDUCATION

Ph.D., Mechanical and Aerospace Engineering — Aug 2024 – Present
The George Washington University — Washington, DC, USA
Advisor: Prof. Peng Wei. Safe multi-agent learning and decision-making.

M.S., Control Science and Engineering — Sep 2021 – May 2024
Tsinghua University, Department of Automation — Beijing, China
Advisor: Prof. Qing Li. Reinforcement learning for fault-tolerant flight control.

B.S., Aircraft Design and Engineering — Sep 2018 – Jun 2021
Beihang University (Beijing University of Aeronautics and Astronautics), School of Aeronautics — Beijing, China
Advisor: Dr. Jidong Wang. Aircraft design, flight dynamics and control, aerodynamics.

Freshman year, Aeronautics and Astronautics — Sep 2017 – Jun 2018
National Cheng Kung University — Tainan, Taiwan

Machine Learning Summer School — Jun 2022 – Aug 2022
University of Oxford — Oxford, United Kingdom

HONORS AND AWARDS

3rd Best Paper of Conference, Best in Track, and Best in Session, DASC 2026 [C3] — 2026
Best in Session Paper Award, ICNS 2026 [C2] — 2026
Graduate Research Assistantship (fully funded Ph.D.), The George Washington University — 2024 – Present
Chinese Government Scholarship (M.S.), Tsinghua University — 2021 – 2024
Chinese Government Scholarship (B.S.), Beihang University — 2018 – 2021
Outstanding Academic Achievement Award, Beihang University — 2019, 2020
Taiwan Ministry of Foreign Affairs Scholarship, NCKU and Fu Jen Catholic University — 2016 – 2018

PUBLICATIONS

All peer reviewed. Project pages, slides and posters at alexzongo.com/publications.

Conference papers

[C7] M. Ghazanfari, M. Casanova, J. Kam, **A. Zongo**, P. Wei, T. Darrell, A. Bayen. *Air Traffic Control Using Large Language Models: Prompt Engineering, Architecture, and Evaluation*. AIAA SciTech Forum, 2027 (to appear). [arXiv] [Project]

[C6] **A. Zongo**, S. Ochoa, L. Sun, A. Nguyen, S. Tran, P. Wei. *Regional Air Mobility Operational Energy Assessment Using a Full-Envelope eVTOL Power Model*. AIAA SciTech Forum, 2027 (to appear).

[C5] **A. Zongo**, F. Fotiadis, U. Topcu, P. Wei. *Robust Multi-Agent Reinforcement Learning for Small UAS Separation Assurance under GPS Degradation and Spoofing*. IEEE Conference on Decision and Control (CDC), 2026 (to appear). Also presented at the ACC 2026 Workshop on Autonomy in Transportation. [arXiv] [Project] [Poster]

[C4] Y. Pang, A. Aziz, **A. Zongo**, P. Wei, K. Crandall. *Large Language Model Guided Reinforcement Learning for Maritime Ship Collision Avoidance*. IEEE OCEANS, Monterey, CA, 2026 (to appear). [Project]

[C3] **A. Zongo**, P. Wei. *Runtime Safety Filtering for Learned Small UAS Separation Policies under GNSS Degradation*. AIAA/IEEE Digital Avionics Systems Conference (DASC), Orlando, FL, 2026 (proceedings forthcoming). 3rd Best Paper of Conference, Best in Track, Best in Session. [arXiv] [Project] [Slides]

[C2] **A. Zongo**, P. Wei. *eVTOL Aircraft Energy Consumption Estimation with Conflict Resolution in High-Density Airspaces*.

Integrated Communications, Navigation and Surveillance Conference (ICNS), Herndon, VA, 2026. Best in Session. [Paper] [arXiv] [Project] [Slides]

[C1] **A. Zongo**, Q. Li. *Towards Intelligent Fault-Tolerant Attitude Flight Control of Fixed-Wing Aircraft*. International Conference on Guidance, Navigation and Control (ICGNC), Changsha, 2024. Lecture Notes in Electrical Engineering, vol. 1353, Springer, 2025. [Paper] [Code] [Slides]

Workshop papers

[W1] I. Sharifi, **A. Zongo**, P. Wei. *Fine-Tuning Large Language Models for Cooperative Tactical Deconfliction of Small Unmanned Aerial Systems*. CVPR Workshop on Multi-Agent Embodied Intelligent Systems (MEIS), Denver, CO, 2026. [Paper] [arXiv] [Project]

Other writing

[E1] **A. Zongo**. *Aircraft Traffic Control: Managing Order in a Crowded Sky*. Research essay, 2025. [Zenodo] [Online]

RESEARCH EXPERIENCE

Graduate Research Assistant Sep 2024 – Present
Intelligent Aerospace Systems Lab, The George Washington University (advisor: Prof. Peng Wei) Washington, DC

- **Robust MARL.** Modeled GNSS degradation and spoofing as a bounded observation adversary and derived a closed-form, first-order worst-case perturbation from the multi-agent value gradient, so robust separation policies train without adversarial co-training [C5].
- **Runtime safety.** Designed and compared runtime filters that wrap a learned separation policy and act on either its action or its observation, using a shared worst-case state estimate [C3].
- **Energy-aware air mobility.** Coupled an eVTOL power model with MVP conflict resolution in BlueSky, measured the energy cost of deconfliction over ∼72,000 simulated transits (median overhead below 1.5%), and learned per-flight reserve bounds with conservative uncertainty estimates [C2, C6]. NASA University Leadership Initiative.
- **Language models.** Co-developed a simulation-to-language pipeline (∼38k rule-consistent samples) and LoRA fine-tuning (SFT, GRPO) of LLMs for sUAS tactical deconfliction; decision accuracy rose from 27% to 88% [W1].

Visiting Researcher May 2026 – Aug 2026
CITRIS and the Banatao Institute, University of California, Berkeley (host: Prof. Alexandre Bayen) Berkeley, CA

- Aircraft trajectory prediction from ADS-B and ATC voice data under uncertain communication: accounting for information lost in transcription and for variability in how an ATC instruction is executed (paper in progress).

Graduate Research Assistant Sep 2021 – May 2024
Navigation and Control Lab, Department of Automation, Tsinghua University (advisor: Prof. Qing Li) Beijing, China

- Developed reinforcement-learning controllers (CEM-TD3, PPO) for fault-tolerant attitude control of a 6-DoF fixed-wing aircraft model under actuator saturation, jamming, and effectiveness loss, wing icing, and a shifted center of gravity; open-sourced the code [C1].

Robotics Software Engineering Intern Jun 2022 – Nov 2022
Popular Robotics Beijing, China

- Built a biped robot simulation in Gazebo with ROS/ROS 2 and designed a curriculum module on gait fundamentals, simulation, and control.

TALKS AND PRESENTATIONS

Embodied AI: Vision-Language-Action Models Oct 2026
Invited panel talk, SEINAR 2026, World AI Week Online

Runtime Safety Filtering for Learned sUAS Separation Policies under GNSS Degradation Sep 2026
Conference talk, DASC 2026 Orlando, FL

Fine-Tuning LLMs for Cooperative Tactical Deconfliction of Small UAS Jun 2026
Poster, MEIS Workshop at CVPR 2026 Denver, CO

Robust MARL for sUAS Separation Assurance under GPS Degradation and Spoofing May 2026
Poster, ACC 2026 Workshop on Autonomy in Transportation New Orleans, LA

eVTOL Energy Overhead Estimation under Conflict Resolution in High-Density Airspaces Apr 2026
Conference talk, ICNS 2026 Herndon, VA

Le parcours d'un Burkinabé dans le domaine de l'aéronautique et de l'espace Apr 2025
Invited webinar, Initiative Burkinabè pour l'Aéronautique et l'Espace (IBAE) Online
Towards Intelligent Fault-Tolerant Attitude Flight Control of Fixed-Wing Aircraft Aug 2024
Conference talk, ICGNC 2024 Changsha, China
Intelligent Control and Reinforcement Learning Apr 2024
Academic forum, Beijing Universities AI Forum and Tsinghua Ph.D. Academic Discussion Beijing, China

TEACHING AND MENTORING

Guest Lecturer, Large Language and Vision Models (MAE 6291 / CSCI 6907) Fall 2026
The George Washington University
Lecture: *From Noise to Data: Diffusion Models through Flow Matching*. [Interactive notes]

Graduate Teaching Assistant, Linear Systems Dynamics (MAE 3134) Spring 2025
The George Washington University
Recitations, office hours, and grading.

Research Mentor Summer 2025 – Present
Intelligent Aerospace Systems Lab, The George Washington University

- M.S. student: maritime traffic simulation and multi-agent RL for collision avoidance (Fall 2025 – present) [C4].
- Undergraduate student: air traffic simulation and research with BlueSky (Summer 2025).

PROFESSIONAL SERVICE

Journal reviewer	IEEE Transactions on Aerospace and Electronic Systems; IEEE Transactions on Intelligent Transportation Systems; Transportation Research Part C; Journal of Guidance, Control, and Dynamics; Journal of Aerospace Information Systems; Engineering Applications of Artificial Intelligence
Conference reviewer	IEEE CDC; IEEE ITSC; ICGNC
Organizing	Safe and Responsible AI Workshop, GWU (Sep 2024), with participants from the FAA, the DOT HASS COE, Johns Hopkins APL, MIT Lincoln Laboratory, TRAILS, NIST, and Responsible AI UK

TECHNICAL SKILLS

Programming	Python, C/C++, Julia, MATLAB/Simulink, ROS/ROS 2, Ada
Machine learning	PyTorch, JAX, NumPy, scikit-learn, OpenCV; LoRA/SFT/GRPO fine-tuning of LLMs
Simulation, design	BlueSky ATM simulator, Gazebo, OpenVSP, SolidWorks, CATIA, ANSYS Fluent
Languages	French (native), English (fluent, C2), Chinese (intermediate, B1)

SELECTED COURSE PROJECTS

- Visual odometry on self-recorded datasets; system identification and nonlinear control of SpaceX Grasshopper dynamics; implementation of Edmonds' blossom algorithm (GWU, 2024–2025).
- Flight simulator of the RCAM aircraft model in MATLAB/Simulink [Code]; conceptual design of a light sport aircraft and preliminary helicopter design (Beihang, 2020–2021).

LEADERSHIP AND OUTREACH

Secretary General, Tsinghua University African Student Association May 2022 – May 2024
R&D Engineer, Tsinghua AI International Student Association Sep 2022 – May 2023
Music and community 2021 – 2024
Church musician, North Cathedral of Beijing; pianist at Tsinghua cultural events; campus volunteer

REFERENCES

Prof. Peng Wei (Ph.D. advisor)
Mechanical and Aerospace Eng., George Washington Univ.
pwei@gwu.edu

Prof. Alexandre Bayen (Berkeley host)
EECS, University of California, Berkeley
bayen@berkeley.edu

Prof. Qing Li (M.S. advisor)
Department of Automation, Tsinghua University
liqing@tsinghua.edu.cn

Prof. Ying Zhao
Computer Science and Technology, Tsinghua University
yingz@tsinghua.edu.cn